

Understanding $U(2)$, $SU(2)$ and Quantum Rotations

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Vipul Bansal

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1 The Foundations of Quantum Matrices

1.1 Definitions of the Continuous Groups

- **$U(2)$ (Unitary Group of Degree 2):** The set of all 2×2 complex matrices U that preserve vector length, satisfying $U^\dagger U = I$. Their determinants lie anywhere on the complex unit circle ($\det(U) = e^{i\phi}$). Every valid single-qubit quantum gate is a $U(2)$ matrix.
- **$SU(2)$ (Special Unitary Group of Degree 2):** A subgroup of $U(2)$ containing matrices with no global phase. They satisfy $U^\dagger U = I$ with the strict added requirement that $\det(U) = 1$. These represent pure physical rotations.
- **$SO(3)$ (Special Orthogonal Group of Degree 3):** The set of 3×3 real matrices representing standard classical 3D rotations, where $R^T R = I$ and $\det(R) = 1$.

1.2 Proof: Decomposing $U(2)$ into $SU(2)$ and Global Phase

Theorem: Any unitary matrix U can be factored into a global phase scalar and a pure $SU(2)$ rotation matrix: $U = e^{i\alpha} S$.

Proof:

1. By definition, $U^\dagger U = I$. Taking the determinant: $\det(U^\dagger) \det(U) = 1 \implies |\det(U)|^2 = 1$. Therefore, $\det(U)$ is a complex phase: $\det(U) = e^{i\theta}$.
2. Define $U = e^{i\alpha} S$. Taking the determinant of both sides (squaring the scalar for a 2×2 matrix):

$$\det(U) = (e^{i\alpha})^2 \det(S) \implies e^{i\theta} = e^{i2\alpha} \det(S)$$

3. To force S to be an $SU(2)$ matrix, we demand $\det(S) = 1$.

$$e^{i\theta} = e^{i2\alpha}(1) \implies \theta = 2\alpha \implies \alpha = \frac{\theta}{2}$$

4. Verify the leftover matrix $S = e^{-i\alpha} U$:

- Unitary check: $S^\dagger S = (e^{-i\alpha}U)^\dagger(e^{-i\alpha}U) = U^\dagger U = I$.
- Determinant check: $\det(S) = (e^{-i\alpha})^2 \det(U) = e^{-i\theta} e^{i\theta} = 1$.

Because S is unitary and $\det(S) = 1$, it is strictly in $SU(2)$.

2 The Algebra of Rotations

2.1 Proof: The Pauli Basis for Traceless Hermitian Unitaries

Theorem: Any 2×2 matrix M that is Hermitian ($M = M^\dagger$), Unitary ($M^\dagger M = I$), and Traceless ($\text{Tr}(M) = 0$) can be written as the dot product of a real 3D unit vector and the Pauli vector: $M = \hat{n} \cdot \vec{\sigma}$.

Proof:

1. **Basis Expansion:** Any 2×2 matrix can be written as $M = c_0 I + c_1 X + c_2 Y + c_3 Z$.
2. **Hermitian Constraint:** If $M = M^\dagger$, the coefficients must be their own complex conjugates, meaning they are strictly real numbers: $n_0, n_1, n_2, n_3 \in \mathbb{R}$.
3. **Traceless Constraint:** $\text{Tr}(M) = 0$. Since Pauli matrices are traceless and $\text{Tr}(I) = 2$:

$$0 = n_0(2) + 0 + 0 + 0 \implies n_0 = 0$$

This leaves $M = n_1 X + n_2 Y + n_3 Z = \vec{n} \cdot \vec{\sigma}$.

4. **Unitary Constraint:** $M^2 = I$. Using the Pauli vector identity $(\vec{a} \cdot \vec{\sigma})^2 = |\vec{a}|^2 I$:

$$M^2 = |\vec{n}|^2 I = I \implies |\vec{n}|^2 = 1$$

Therefore, $\vec{n}$ is a 3D real unit vector $\hat{n}$, and $M = \hat{n} \cdot \vec{\sigma}$.

2.2 Proof: The Matrix Exponential and Generators of Rotation

Theorem: A continuous 3D rotation by angle ϕ around axis $\hat{n}$ is generated by the exponential of the Pauli matrices: $R_{\hat{n}}(\phi) = \cos\left(\frac{\phi}{2}\right) I - i \sin\left(\frac{\phi}{2}\right) (\hat{n} \cdot \vec{\sigma})$.

Proof:

1. **Infinitesimal Limit:** An infinitesimally small unitary transformation is $U(d\phi) = I - iGd\phi$. To rotate around $\hat{n}$, the Hermitian generator is $G = \frac{1}{2}(\hat{n} \cdot \vec{\sigma})$. Extending this to a continuous angle ϕ using the limit definition of Euler's exponential yields $e^{-i\frac{\phi}{2}(\hat{n} \cdot \vec{\sigma})}$.
2. **Maclaurin Series Expansion:**

$$e^{-i\frac{\phi}{2}(\hat{n} \cdot \vec{\sigma})} = I - i\frac{\phi}{2}(\hat{n} \cdot \vec{\sigma}) - \frac{1}{2!} \left(\frac{\phi}{2}\right)^2 (\hat{n} \cdot \vec{\sigma})^2 + \dots$$

3. Apply the identity $(\hat{n} \cdot \vec{\sigma})^2 = I$. Group the even powers (containing I) and odd powers (containing $-i(\hat{n} \cdot \vec{\sigma})$):

$$I \left[1 - \frac{1}{2!} \left(\frac{\phi}{2}\right)^2 + \dots \right] - i(\hat{n} \cdot \vec{\sigma}) \left[\left(\frac{\phi}{2}\right) - \frac{1}{3!} \left(\frac{\phi}{2}\right)^3 + \dots \right]$$

4. These brackets are the exact infinite Taylor series for cosine and sine, yielding the closed-form representation.

3 The Geometric Mapping (The Double Cover)

3.1 Proof: Why Every $SU(2)$ Matrix is a Rotation Matrix

Theorem: Every $SU(2)$ matrix is structurally forced to take the exact trigonometric form of a 3D spatial rotation because it maps to a 4D hypersphere.

Proof:

1. An arbitrary $SU(2)$ matrix, constrained by $U^\dagger U = I$ and $\det(U) = 1$, strictly takes the form:

$$U = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix} \quad \text{where} \quad |a|^2 + |b|^2 = 1$$

2. Decompose complex components a and b into four real numbers: $a = x_0 - ix_3$ and $b = -x_2 - ix_1$.

3. Substitute into the determinant condition:

$$x_0^2 + x_1^2 + x_2^2 + x_3^2 = 1$$

This is the exact geometric equation of a 4D hypersphere.

4. Define $x_0 = \cos\left(\frac{\phi}{2}\right)$. Using Pythagorean identities ($1 - \cos^2 \theta = \sin^2 \theta$), the remaining three variables must have a total length of $\sin\left(\frac{\phi}{2}\right)$. They can be factored into a pure 3D unit vector $\hat{n} = (n_x, n_y, n_z)$:

$$\begin{aligned} x_1 &= n_x \sin\left(\frac{\phi}{2}\right) \\ x_2 &= n_y \sin\left(\frac{\phi}{2}\right) \\ x_3 &= n_z \sin\left(\frac{\phi}{2}\right) \end{aligned}$$

5. Rebuild the original matrix with these definitions:

$$U = \begin{pmatrix} \cos\left(\frac{\phi}{2}\right) - in_z \sin\left(\frac{\phi}{2}\right) & -n_y \sin\left(\frac{\phi}{2}\right) - in_x \sin\left(\frac{\phi}{2}\right) \\ n_y \sin\left(\frac{\phi}{2}\right) - in_x \sin\left(\frac{\phi}{2}\right) & \cos\left(\frac{\phi}{2}\right) + in_z \sin\left(\frac{\phi}{2}\right) \end{pmatrix}$$

6. Separating the identity matrix from the Pauli matrix components yields exactly:

$$U = \cos\left(\frac{\phi}{2}\right) I - i \sin\left(\frac{\phi}{2}\right) (\hat{n} \cdot \vec{\sigma})$$

3.2 The Physics of the Double Cover ($SU(2) \rightarrow SO(3)$)

Because of the $\phi/2$ in the rotation formula, rotating a quantum state by 360° (2π) multiplies the state by -1 . A full 720° (4π) rotation is required to return to the identity matrix.

This mathematical divergence explains why identical physical rotations can yield mathematically opposite axes. For a 180° rotation ($\theta = \pi$), evaluating the global phase as $\pi/2$ or $-\pi/2$ results in $\hat{n}$ and $-\hat{n}$, respectively. Physically, rotating 180° clockwise puts a system in the exact same orientation as rotating 180° counter-clockwise.

4 Practical Gate Computations & Algorithmic Implementation

4.1 The S Gate Phase Relation

The standard S gate is a $U(2)$ matrix that acts physically as a $\pi/2$ rotation around the Z-axis, but requires a global phase correction to match the pure $SU(2)$ rotation matrix $R_z(\pi/2)$:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = e^{i\pi/4} R_z(\pi/2)$$

4.2 Algorithm: Extracting $(\alpha, \hat{n}, \theta)$ from an arbitrary $U(2)$ matrix

To computationally reverse-engineer the Bloch sphere parameters from a 2×2 unitary matrix $\mathcal{U}$:

1. **Extract the Global Phase (α):** Extract the phase using the mathematically stable arctangent representation of the determinant.

$$\alpha = \frac{1}{2} \arg(\det(\mathcal{U}))$$

2. **Isolate the $SU(2)$ Rotation Matrix ($\tilde{g}$):** Remove the global phase.

$$\tilde{g} = e^{-i\alpha} \mathcal{U}$$

3. **Extract the Rotation Angle (θ):** Using $\text{Tr}(X, Y, Z) = 0$, the trace of the rotation matrix isolates the cosine term.

$$\theta = 2 \arccos \left(\frac{\text{Tr}(\tilde{g})}{2} \right)$$

4. **Extract the 3D Axis ($\hat{n}$):** Multiply the matrix by each Pauli matrix (σ_j) and take the trace. Due to orthogonality ($\text{Tr}(\sigma_j \sigma_k) = 2\delta_{jk}$), this isolates coordinate n_j .

$$n_j = \frac{i \cdot \text{Tr}(\sigma_j \tilde{g})}{2 \sin \left(\frac{\theta}{2} \right)}$$